

Preface

Topology

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Chapter 1

Topological Spaces

§ 1.1. Topological Spaces

Definition 1.1.1. A topology on a set X is a collection T of subsets of X such that:

- (1) $\emptyset, X \in T$;
- (2) if $\{U_i\}_{i \in I} \subseteq T$, then $\bigcup_{i \in I} U_i \in T$;
- (3) if $\{U_i\}_{i=1}^n \subseteq T$, then $\bigcap_{i=1}^n U_i \in T$.

We call the pair (X, T) a topological space.

Example 1.1.1. Let $X = \{a, b, c\}$. Then: $T_1 := \{\emptyset, \{a, b, c\}, \{b\}, \{a, b\}, \{b, c\}\}$ and $T_2 := \{\emptyset, \{b\}, \{a, b, c\}\}$ are topologies on X . However, $T_3 = \{\emptyset, \{a\}, \{b\}, \{a, b, c\}\}$ is *not* a topology on X .

Example 1.1.2 (Every set admits a topology). Let X be a set. The *discrete topology* on X is $T = \mathcal{P}(X)$. The *trivial topology* on X is $T = \{\emptyset, X\}$.

Definition 1.1.2. Let T_1 and T_2 be topologies on a set X . If $T_1 \subseteq T_2$, then we say T_2 is finer than T_1 , and likewise that T_1 is coarser than T_2 .

Example 1.1.3. Let $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$ be vectors in $\mathbb{R}^n$. Define $d(\vec{x}, \vec{y}) := \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$. This is a metric on $\mathbb{R}^n$, and we will see this induces a topology on $\mathbb{R}^n$.

Definition 1.1.3. Let (X, d) be a metric space, $x \in X$, and $\epsilon > 0$. The open ball centered at x with radius ϵ is denoted $B(x, \epsilon) = \{y \in X \mid d(x, y) < \epsilon\}$.

Definition 1.1.4. Let (X, d) be a metric space. A set $U \subseteq X$ is open if for all $x \in U$, there exists $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$. The collection of open sets of X is denoted τ_X .

Proposition 1.1.1. Let (X, d) be a metric space. Then (X, τ_X) is a topological space.

Proof. Clearly $\emptyset, X \in \tau_X$. Let $\{U_i\}_{i \in I} \subseteq \tau_X$ be an arbitrary family of open sets. Let $x \in \bigcup_{i \in I} U_i$. Then $x \in U_i$ for some i . So there exists an $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U_i \subseteq \bigcup_{i \in I} U_i$. Now let $\{U_i\}_{i=1}^n$ be a finite collection of open sets. Let $x \in \bigcap_{i=1}^n U_i$. Then $x \in U_i$ for

all $i = 1, \dots, n$. So there exists $\epsilon_i > 0$ such that $B(x, \epsilon_i) \subseteq U_i$ for all $i = 1, \dots, n$. Define $\epsilon := \min\{\epsilon_1, \dots, \epsilon_n\}$. Then $B(x, \epsilon) \subseteq U_i$ for all $i = 1, \dots, n$. Hence $B(x, \epsilon) \subseteq \bigcap_{i=1}^n U_i$. Thus (X, τ_X) is a topological space. $\square$

Example 1.1.4. Using the same terminology from Example 1.1.3, since $d(\vec{x}, \vec{y})$ is a metric, $(\mathbb{R}^n, \tau_{\mathbb{R}^n})$ is a topological space.

§ 1.2. Topologies Generated by a Basis

Definition 1.2.1. Let X be a set. A basis for a topology on X is a collection $\mathcal{B}$ of subsets of X such that:

- (1) for all $x \in X$, there exists $B \in \mathcal{B}$ such that $x \in B$.
- (2) If $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Remark. Let (X, d) be a metric space and $\mathcal{B}$ a basis for a topology on X . It follows from the above definition that for any $U \in \tau_X$, there exists an open basis element $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Example 1.2.1. The sets $\mathcal{B}_1 := \{(a, b) \mid a < b\}$ and $\mathcal{B}_2 := \{(a, b) \mid a < b, a, b \in \mathbb{Q}\}$ are bases for a topology on $\mathbb{R}$. These definitions extend nicely to $\mathbb{R}^n$ by considering open balls.

Example 1.2.2 (Every set admits a basis for a topology). Let X be a set. Then $\mathcal{B} := \{\{p\} \mid p \in X\}$ is a basis for a topology on X .

Proposition 1.2.1. Let X be a set and $\mathcal{B}$ a basis for a topology on X . Let τ be the set containing the empty set and every subset of X which is equal to a union of basis elements of $\mathcal{B}$. Then (X, τ) is a topological space.

Proof. $\square$

Definition 1.2.2. Let X be a set and $\mathcal{B}$ a topology on X . The topology defined in Proposition 1.2.1 is called the topology generated by $\mathcal{B}$.

Example 1.2.3. From Example 1.2.1, $\mathcal{B}_1$ and $\mathcal{B}_2$ generate the standard topology on $\mathbb{R}^n$. From Example 1.2.2, $\mathcal{B}$ generates the discrete topology on X .

§ 1.3. Product Topologies

Definition 1.3.1. Let (X, τ_1) and (Y, τ_2) be topological spaces. The product topology on $X \times Y$ is the topology generated by the basis $\mathcal{B} := \{U \times V \mid U \in \tau_1, V \in \tau_2\}$.

Remark. Note that $\mathcal{B}$ is not necessarily a topology on $X \times Y$. Indeed, consider $X = Y = \{0, 1\}$ in the discrete topology. Then the sets $\{0\} \times \{0\}$ and $\{0\} \times \{1\}$ are in $\mathcal{B}$ as per the above definition, but their union $\{(0, 0), (0, 1)\}$ is not. Hence $\mathcal{B}$ is not a topology.

Example 1.3.1. Recall from Example 1.2.1 that $\mathcal{B} := \{(a, b) \mid a < b\}$ generates the standard topology on $\mathbb{R}$. Consider the following to bases for a topology on $\mathbb{R}^2$:

$$\mathcal{B}_1 := \{B(x, r) \mid x \in \mathbb{R}^2, r > 0\},$$

$$\mathcal{B}_2 := \{I_1 \times I_2 \mid I_1, I_2 \in \mathcal{B}\}.$$

The basis elements of $\mathcal{B}_1$ are open balls of radius r , whilst the basis elements of $\mathcal{B}_2$ are open squares. Note that the basis $\mathcal{B}_2$ interprets $\mathbb{R}^2$ as the product $\mathbb{R} \times \mathbb{R}$.

This idea can be further generalized to $\mathbb{R}^3$. Consider:

$$\mathcal{B}_3 := \{B(x, r) \mid x \in \mathbb{R}^3, r > 0\}$$

$$\mathcal{B}_4 := \{I_1 \times I_2 \times I_3 \mid I_1, I_2, I_3 \in \mathcal{B}\}$$

$$\mathcal{B}_5 := \{B \times I \mid B \in \mathcal{B}_1, I \in \mathcal{B}\}.$$

The basis $\mathcal{B}_4$ interprets $\mathbb{R}^3$ as $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$, whilst the basis $\mathcal{B}_5$ interprets $\mathbb{R}^3$ as $\mathbb{R}^2 \times \mathbb{R}$.

Proposition 1.3.1. *The bases $\mathcal{B}_1$ and $\mathcal{B}_2$ from Example 1.3.1 generate the same topologies on $\mathbb{R}^2$.*

Proof.

□

§ 1.4. Subspace Topology

Proposition 1.4.1. *Let (X, T) be a topological space. If $Y \subseteq X$, then $T_Y := \{Y \cap U \mid U \in T_X\}$ is a topology on Y .*

Proof.

□

Definition 1.4.1. Let (X, T) be a topological space and $Y \subseteq X$. The *subspace topology on Y* is the topology defined in Proposition 1.4.1. We call Y a subspace of X if it is paired with the subspace topology.

Example 1.4.1. Consider the interval $[2, 4)$ with the subspace topology inherited from $\mathbb{R}$. Note that $(3, 4)$ is open in $[2, 4)$ and $\mathbb{R}$, while $[2, 3)$ is open in $[2, 4)$ but *not* open in $\mathbb{R}$.

Example 1.4.2. Consider the integers $\mathbb{Z}$ with the subspace topology inherited from $\mathbb{R}$. For any $n \in \mathbb{Z}$, we have $\{n\} = \mathbb{Z} \cap (n - \frac{1}{2}, n + \frac{1}{2})$. Since $(n - \frac{1}{2}, n + \frac{1}{2})$ is open in $\mathbb{R}$, each singleton in $\mathbb{Z}$ is open. Hence the subspace topology on $\mathbb{Z}$ inherited from $\mathbb{R}$ is the discrete topology.

Proposition 1.4.2. If $\mathcal{B}$ is a basis for a topology on a set X , and Y is a subset of X , then $\mathcal{B}_Y := \{Y \cap U \mid U \in \mathcal{B}\}$ is a basis for Y .

Proof.

□

Example 1.4.3. Recall that $\mathcal{B} := \{(a, b) \mid a < b\}$ is a basis for the standard topology on $\mathbb{R}$. Consider $\mathcal{B}_{\mathbb{Q}} := \{(a, b) \cap \mathbb{Q} \mid a < b\}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, we have that $(a, b) \cap \mathbb{Q} \neq \emptyset$. In fact, each nonempty open set of $\mathbb{Q}$ with this particular subspace topology is infinite, hence it is not the discrete topology.

§ 1.5. Closed Sets and Limit Points

Definition 1.5.1. A subset A of a topological space X is closed in X if A^c is open.

Example 1.5.1. Let (X, T) be a topological space. Then $\emptyset$ and X are closed.

Example 1.5.2. If $x \in \mathbb{R}^n$, then $\{x\}$ is closed in $\mathbb{R}^n$. Indeed, for any $p \in \mathbb{R}^n \setminus \{x\}$, we have that $p \in B(p, \frac{\|p-x\|}{3}) \subseteq \mathbb{R}^n$.

Example 1.5.3. Given any $x \in \mathbb{R}^n$, the set $\{y \in \mathbb{R}^n \mid \|x - y\| \leq \epsilon\}$ is closed. The proof of this fact follows identically to Example 1.5.2.

Example 1.5.4. Consider the topological space $(X, \mathcal{P}(X))$. Since every set is open, every set must be closed.

Proposition 1.5.1. Let X be a topological space. Then:

- (1) $\emptyset$ and X are closed;
- (2) If $\{C_i\}_{i \in I}$ is an arbitrary family of closed sets, then $\bigcap_{i \in I} C_i$ is closed;
- (3) If $\{C_i\}_{i=1}^n$ is a finite collection of closed sets, then $\bigcup_{i=1}^n C_i$ is closed.

Proof. This follows from De Morgan's laws and Definition 1.1.1.

□

Example 1.5.5. Every finite subset of $\mathbb{R}^n$ is closed. Indeed, any finite subset can be written as the finite union of singletons, which are closed in $\mathbb{R}^n$ by Example 1.5.2.

Proposition 1.5.2. *Let Y be a subspace of a topological space X . Then a set $A \subseteq Y$ is closed in Y if and only if $A = Y \cap C$, where C is a closed set in X .*

Proof.

□

§ 1.6. Closure and Interior

Definition 1.6.1. Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$.

- (1) The interior of A is $A^\circ := \bigcup \{U \mid U \subseteq A, U \text{ open}\}$.
- (2) The closure of A is $\bar{A} := \bigcap \{C \mid C \supseteq A, C \text{ closed}\}$.

Remark.

- (1) Clearly $A^\circ \subseteq A \subseteq \bar{A}$. The interior of any set is open and is the largest open subset containing it. The closure of any set is closed and is the smallest closed subset contained in it. A set is open if and only if it is equal to its interior. A set is closed if and only if it is equal to its closure. If C is any closed set in X and $A \subseteq C$, then $\bar{A} \subseteq C$.

Example 1.6.1. If $F \subseteq \mathbb{R}^n$ is finite then $F^\circ = \emptyset$ and $\bar{F} = F$.

Example 1.6.2. Consider the interval $(2, 3]$ in $\mathbb{R}$. Then $(2, 3]^\circ = (2, 3)$ and $\overline{(2, 3]} = [2, 3]$.

Example 1.6.3. In the discrete topology, every set is open and closed, hence every set equals its interior and closure.

Example 1.6.4. Consider the set $A := \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$ as a subset of $\mathbb{R}$. Then $A^\circ = \emptyset$ and $\bar{A} = A \cup \{0\}$.

Proposition 1.6.1. *Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$.*

- (1) $p \in \bar{A}$ if and only if for all open sets U in X containing p , we have $U \cap A \neq \emptyset$.
- (2) If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $p \in \bar{A}$ if and only if for all basis elements $B \in \mathcal{B}$ containing p , we have $B \cap A \neq \emptyset$.

Proof.

□

§ 1.7. Limit Points of a Set

Definition 1.7.1. Let (X, T) be a topological space. Let $A \subseteq X$. A point $p \in X$ is a limit point of A if $p \in \overline{A} \setminus \{p\}$.

Example 1.7.1. Consider $\mathbb{R}$ with the standard topology.

- (1) If F is a finite set, then the set containing the limit points of F is empty.
- (2) The set containing the limit points of $(2, 3]$ is $[2, 3]$.
- (3) The set containing the limit points of $\mathbb{Q}$ is $\mathbb{R}$.
- (4) The set containing the limit points of $\{\frac{1}{n} \mid n \in \mathbb{N}\}$ is $\{0\}$.

Example 1.7.2. Let X be a set with the discrete topology. For any $A \subseteq X$, the set containing the limit points of A is empty.

Proposition 1.7.1. Let (X, T) be a topological space. Let $A \subseteq X$. Then A is closed in X if and only if A contains all of its limit points.

Proof.

$$\begin{aligned}
 A \text{ is closed in } X &\iff A^c \text{ is open in } X \\
 &\iff (\forall p \in A^c)(\exists U \in T) : p \in U \subseteq A^c \\
 &\iff (\forall p \in A^c)(\exists U \in T) : p \in U \text{ and } U \cap A = \emptyset \\
 &\iff (\forall p \in A^c) : p \notin \overline{A} \\
 &\iff (\forall p \in A^c) : p \notin \overline{A \setminus \{p\}} \\
 &\iff (\forall p \in A^c) : p \notin \overline{A \setminus \{p\}} \\
 &\iff A \text{ contains its limit points.}
 \end{aligned}$$

□

Proposition 1.7.2. Let (X, T) be a topological space and $A \subseteq X$. Then:

$$\overline{A} = A \cup \{p \mid p \text{ is a limit point of } A\}.$$

Proof. Let $p \in \overline{A}$. If $p \in A$, then by Proposition 1.7.1 we are done. Suppose $p \in \overline{A} \setminus A$. Since $p \in \overline{A}$, for all open sets $U \subseteq X$ containing p , we have $U \cap A \neq \emptyset$. But since $p \notin A$, this is equivalent to the statement that for all open sets $U \subseteq X$ containing p , we have $U \cap A \setminus \{p\} \neq \emptyset$. Therefore $p \in \overline{A \setminus \{p\}}$. This establishes $\overline{A} \subseteq A \cup \{p \mid p \text{ is a limit point of } A\}$.

Now let $p \in A \cup \{p \mid p \text{ is a limit point of } A\}$. If $p \in A$, then $p \in \overline{A}$. If $p \in \{p \mid p \text{ is a limit point of } A\}$, then $p \in \overline{A}$ by Proposition 1.7.1. Hence $\overline{A} = A \cup \{p \mid p \text{ is a limit point of } A\}$. □

§ 1.8. Hausdorff Spaces

Definition 1.8.1. Let $(X, \mathcal{T})$ be a topological space. We say X is *Hausdorff* (or T_2) if for all $x, y \in X$, there exist an open set $U \subseteq X$ containing x and an open set $V \subseteq X$ containing y such that $U \cap V = \emptyset$.

Example 1.8.1. $\mathbb{R}^n$ is Hausdorff. Let $x, y \in \mathbb{R}^n$. Let $\epsilon = \frac{\|x-y\|}{3}$. Then $B(x, \epsilon) \cap B(y, \epsilon) = \emptyset$.

Example 1.8.2. Consider a set X with the discrete topology. Then X is Hausdorff since for any $x, y \in X$, $\{x\} \cap \{y\} = \emptyset$.

Proposition 1.8.1. *In any Hausdorff space, every finite set is closed.*

Proof. Let $p, q \in X$. Then there exists an open set U containing p and an open set V containing q such that $U \cap V = \emptyset$. Now $p \in X \setminus \{q\}$. Hence $p \in U \subseteq X \setminus \{q\}$. Therefore $X \setminus \{q\}$ is open, establishing that $\{q\}$ is closed. Since any finite set can be written as the finite union of singletons, every finite set in a Hausdorff space is closed. $\square$

Proposition 1.8.2. *Let $(X, \mathcal{T})$ be Hausdorff and $A \subseteq X$. Then a point $p \in X$ is a limit point of A if and only if every open set $U \subseteq X$ containing p contains infinitely many points of A .*

Proof.

$\square$

§ 1.9. Continuity

Definition 1.9.1. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. A function $f : X \rightarrow Y$ is *continuous* if for all $U \in \mathcal{T}_Y$, we have $f^{-1}(U) \in \mathcal{T}_X$.

Example 1.9.1. Let X be a set with the discrete topology and $(Y, \mathcal{T})$ any topological space. Then any map $f : X \rightarrow Y$ is continuous.

Proposition 1.9.1. *Let X, Y be topological spaces. Let $f : X \rightarrow Y$ be a map. The following are equivalent:*

- (1) f is continuous;
- (2) For all $A \subseteq X$, $f(A) \subseteq \overline{f(A)}$;
- (3) $f^{-1}(C)$ is closed in X for any C closed in Y .
- (4) For all $x_0 \in X$ and for all $V \in \mathcal{T}_Y$ such that $f(x_0) \in V$, there exists a $U \in \mathcal{T}_X$ such that $x_0 \in U$ and $f(U) \subseteq V$.

Proof.

□

§ 1.10. Metric Topologies

Definition 1.10.1. Let X be a set. A metric is a map $d : X \times X \rightarrow \mathbb{R}$ satisfying for all $x, y, z \in X$:

- (1) $d(x, x) = 0$;
- (2) if $d(x, y) = 0$, then $x = y$;
- (3) $d(x, y) = d(y, z)$;
- (4) $d(x, y) \leq d(x, z) + d(z, y)$;

We call the pair (X, d) a metric space.

Definition 1.10.2. Let (X, d) be a metric space.

- (1) The open ball centered around $x_0 \in X$ with radius $r > 0$ is denoted:

$$B(x_0, r) := \{x \in X \mid d(x, x_0) < r\}.$$

- (2) The closed ball centered around $x_0 \in X$ with radius $r > 0$ is denoted:

$$\bar{B}(x_0, r) := \{x \in X \mid d(x, x_0) \leq r\}.$$

- (3) The sphere centered around $x_0 \in X$ with radius $r > 0$ is denoted:

$$S(x_0, r) := \{x \in X \mid d(x, x_0) = r\}.$$

Definition 1.10.3. A metric topology is the topology generated by the basis of open balls defined by the given metric.

Definition 1.10.4. Let (X, T) be a topological space. We say X is metrizable if there exists a metric d that induces the topology on X .

Example 1.10.1. Consider the discrete topology $(X, \mathcal{P}(X))$. Then X is metrizable. Indeed, define $d : X \times X \rightarrow \mathbb{R}$ by:

$$d(x, y) = \begin{cases} 1, & x \neq y \\ 0, & x = y \end{cases}.$$

This is a metric, and is called the *discrete metric*. Let T_d denote the topology generated by the open balls with respect to the metric d . Let $Y \subseteq X$ be open. Then $Y = \bigcup_{y \in Y} \{y\}$. But note that we can write $\{y\} = B_d(y, \frac{1}{2})$. Hence $Y = \bigcup_{y \in Y} B_d(y, \frac{1}{2})$. This gives the inclusion $\mathcal{P}(X) \subseteq T_d$. Conversely, since every open ball is a subset of X , surely it is open in the discrete topology. Hence $T_d \subseteq \mathcal{P}(X)$.

missing a lot of stuff on metric topologies

§ 1.11. Quotient Topologies

Definition 1.11.1. Let X, Y be topological spaces and $p : X \rightarrow Y$ a continuous surjection. We say p is a quotient map if U is open in Y if and only if $p^{-1}(U)$ is open in X .

Example 1.11.1. Let T denote the standard topology on $\mathbb{R}^n$. Consider the map $p : (\mathbb{R}, \mathcal{P}(\mathbb{R})) \rightarrow (\mathbb{R}, T)$ given by $p(x) = x$. This is not a quotient map since $p^{-1}(\{3\})$ is open in $(\mathbb{R}, \mathcal{P}(\mathbb{R}))$ but is not open in $(\mathbb{R}, T)$.

Proposition 1.11.1. Let (X, T) be a topological space and Y a set. Let $p : X \rightarrow Y$ be a surjection. The set $\{U \subseteq Y \mid p^{-1}(U) \text{ open in } X\}$ is the unique topology on Y such that p is a quotient map.

Definition 1.11.2. The topology in Proposition 1.11.1 is called the quotient topology on Y induced by p .

Proposition 1.11.2. Let (X, T) be a topological space and Y a set. Let $p : X \rightarrow Y$ be a surjection. The quotient topology on Y induced by p is the finest topology on Y such that p is continuous.

Definition 1.11.3. Let $\sim$ be an equivalence relation on a topological space X . Let $X/\sim$ denote the set of distinct equivalence classes. Let $p : X \rightarrow X/\sim$ be the natural surjection given by $p(x) = [x]$. Let $T_\sim$ be the quotient topology on Y induced by p . We call the space $(X/\sim, T_\sim)$ the quotient space of X modulo $\sim$.

Example 1.11.2. Consider the set $S := \{(x, y) \mid x \in [0, 1], y \in [0, 1]\}$. Define $\sim$ on S by:

$$\begin{aligned} (0, y) &\sim (1, y) \text{ for all } y \in [0, 1] \\ (x, y) &\sim (x, y) \text{ for all } (x, y) \in (0, 1) \times [0, 1]. \end{aligned}$$

This is an equivalence relation. The space $(S/\sim, T_\sim)$ is homeomorphic to a cylinder in $\mathbb{R}^3$.

Example 1.11.3. Let $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$. Define $\sim$ on D by:

$$\begin{aligned} (x, y) &\sim (x, y) \text{ whenever } x^2 + y^2 < 1, \\ (x, y) &\sim (a, b) \text{ whenever } x^2 + y^2 = 1. \end{aligned}$$

This is an equivalence relation. The space $(D/\sim, T_\sim)$ is homeomorphic to a sphere in $\mathbb{R}^3$.

stuff on group actions

Corollary 1.11.1. *Let $g : X \rightarrow Z$ be a surjective continuous map. Let $X^* = \{g^{-1}(\{z\}) \mid z \in Z\}$. Give X^* the quotient topology. Then g induces a bijective continuous map $f : X^* \rightarrow Z$ which is a homeomorphism if and only if g is a quotient map.*

Proof.

□

Example 1.11.4. Define $\sim$ on $[0, 1] \times [0, 1]$ by:

$$(0, y) \sim (1, y) \text{ for all } y \in [0, 1]$$

$$(x, 0) \sim (x, 1) \text{ for all } x \in [0, 1]$$

$$(x, y) \sim (x, y) \text{ for all } (x, y) \in (0, 1) \times (0, 1).$$

We will prove that $([0, 1] \times [0, 1])/\sim$ is homeomorphic to $S^1 \times S^1$. Define $g : [0, 1] \times [0, 1] \rightarrow S^1 \times S^1$ by $g(s, t) = (\cos 2\pi s, \sin 2\pi s, \cos 2\pi t, \sin 2\pi t)$. This map is continuous and surjective.

Chapter 2

Connectedness and Compactness

§ 2.1. Connected Spaces

Definition 2.1.1. Let X be a topological space.

- (1) A separation of X is a pair U, V of disjoint and nonempty open subsets of X whose union is X . If a separation of X exists, we say X is disconnected.
- (2) The space X is connected if there does not exist a separation of X .

Proposition 2.1.1. A nonempty topological space X is connected if and only if the only subsets of X that are both open and closed are $\emptyset$ and X .

Proof. Suppose X is connected. Note that $\emptyset, X$ are by definition open and closed. If there were another subset A such that $A^c = A$, then $X = A \cup A^c$ would be a separation. This is a contradiction, hence $\emptyset, X$ are the only open and closed subsets of X .

For the converse, we consider the contrapositive of the above statement. If X is a disconnected topological space, then there exists nonempty open subsets U, V of X such that $U \cap V = \emptyset$ and $U \cup V = X$. Then U and V are closed and open, as $U^c = V$ and $V^c = U$. □

Example 2.1.1. Consider $[0, 1] \cup [2, 3] \subseteq \mathbb{R}$ with the subspace topology. Then $[0, 1] \cup [2, 3]$ is not connected since $U = [0, 1]$ and $V = [2, 3]$ form a separation.

Example 2.1.2. Consider the topological space $(X, \mathcal{P}(X))$. If X has at least two points, then X is not connected, as $\{p\}, X \setminus \{p\}$ form a separation.

Lemma 2.1.1. Let Y be a subspace of a topological space X .

- (1) A, B is a separation of Y ;
- (2) $Y = A \cup B$, where $A, B \subseteq Y$ are disjoint, nonempty, and $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.

Proof. Let A, B be a separation of Y . Then $Y = A \cup B$, where $A, B \subseteq Y$ are disjoint, nonempty, and open in Y . We only need to show that $\overline{A} \cap B = A \cap \overline{B} = \emptyset$. Suppose towards

contradiction we can find some $p \in \bar{A} \cap B$. Since B is open in Y , we can write $B = Y \cap U$, where $U \subseteq X$ is open. Hence $p \in \bar{A} \cap Y \cap U$. Now since $p \in \bar{A}$ and $p \in U$, the set $A \cap U$ is nonempty. Hence we can find some $a \in A \cap U$. But since $A \subseteq Y$, we have $a \in Y \cap U = B$. Thus $a \in A \cap B$, which is a contradiction. By a similar argument, $A \cap \bar{B} = \emptyset$.

Conversely, we must show that A and B are open in Y . Since $\bar{A} \cap B = \emptyset$, note that:

$$\begin{aligned}\bar{A} \cap Y &= \bar{A} \cap (A \cup B) \\ &= (\bar{A} \cap A) \cup (\bar{A} \cap B) \\ &= A \cup \emptyset \\ &= A.\end{aligned}$$

A similar argument will show that $\bar{B} \cap Y = B$. Now since $\bar{A}$ and $\bar{B}$ are closed, by definition this means A and B are closed in Y . Since $A = Y \setminus B$ and $B = Y \setminus A$, they are open in Y as well. $\square$

Example 2.1.3. Consider the following subset of $\mathbb{R}^2$:

$$\{(x, 0) \mid x \in \mathbb{R}\} \cup \{(x, \frac{1}{x}) \mid x > 0\}$$

This set is not connected, as neither set contains a limit point of the other

Lemma 2.1.2. *If C and D form a separation of X , and if Y is a connected subspace of X , then either $Y \subseteq C$ or $Y \subseteq D$.*

Proof. Suppose towards contradiction $Y \cap C \neq \emptyset$ and $Y \cap D \neq \emptyset$. Since C and D are open in X , then $Y \cap C$ and $Y \cap D$ are open in Y . Moreover, their union is Y and their intersection is empty. Therefore, $Y \cap C$ and $Y \cap D$ form a separation of Y , which is a contradiction. $\square$

Theorem 2.1.1. *The union of a collection of connected subspaces of X that have a point in common is connected.*

Proof. Let $\{A_i\}_{i \in I}$ be a collection of connected subspaces of X and let $p \in \bigcap_{i \in I} A_i$. Suppose towards contradiction that U, V is a separation of $\bigcup_{i \in I} A_i$. Then either $p \in U$ or $p \in V$. Without loss of generality, assume $p \in U$. Since each A_i is connected, it must lie entirely in either U or V , and it cannot lie in V because it contains the point $p \in U$. Hence $A_i \subset U$ for all i , which means $\bigcup_{i \in I} A_i \subset U$. But this means that $V = \emptyset$, which is a contradiction. $\square$

Theorem 2.1.2. *Let A be a connected subspace of X . If $A \subseteq B \subseteq \bar{A}$, then B is connected.*

Proof. Suppose towards contradiction $B = C \cup D$ is a separation of B . Then $A \subseteq C$ or $A \subseteq D$; suppose without loss of generality that $A \subseteq C$. Then $\bar{A} \subseteq \bar{C}$. By Lemma 2.1.1,

$\overline{C} \cap D = \emptyset$. Since $B \subseteq \overline{A} \subseteq \overline{C}$, B cannot intersect D . This contradicts D being a nonempty subset of B . $\square$

Example 2.1.4. The *Topologist's sine curve* is the set:

$$S := \{(x, \sin \frac{1}{x}) \mid 0 < x \leq 1\} \cup \{(0, y) \mid y \in [-1, 1]\}.$$

Note that the closure of $\{(x, \sin \frac{1}{x}) \mid 0 < x \leq 1\}$ is S . Since $\{(x, \sin \frac{1}{x}) \mid 0 < x \leq 1\}$ is connected, S is connected.

Theorem 2.1.3. *The image of a connected space under a continuous function is connected.*

Proof. Let X be connected and $f : X \rightarrow Y$ continuous. If U and V form a separation of $f(X)$, then $f^{-1}(U)$ and $f^{-1}(V)$ form a separation of X , which is a contradiction. $\square$

Theorem 2.1.4. *A finite Cartesian product of connected spaces is connected*

Proof. We will only prove the theorem for the product of two connected spaces X and Y , as the finite case follows by induction. Choose a point $(a, b) \in X \times Y$. The set $X \times \{b\}$ is connected since it is homeomorphic to X . For each $x_0 \in X$, the set $\{x_0\} \times Y$ is connected since it is homeomorphic to Y . Let $T_{x_0} := (X \times \{b\}) \cup (\{x_0\} \times Y)$. Since $(x_0, b) \in (X \times \{b\}) \cap (\{x_0\} \times Y)$, by Theorem 2.1.1 T_{x_0} is connected. Now $\bigcup_{x_0 \in X} T_{x_0}$ is connected, as $(a, b) \in \bigcap_{x_0 \in X} T_{x_0}$. But $\bigcup_{x_0 \in X} T_{x_0} = X \times Y$, hence $X \times Y$ is connected. $\square$

§ 2.2. Connected Subspaces of $\mathbb{R}$

Theorem 2.2.1. *A set $E \subseteq \mathbb{R}$ is connected if and only if whenever $a < c < b$ with $a, b \in E$, then $c \in E$.*